

PRANJALI HIMANSHU SANGHVI

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EDUCATION

San Jose State University, San Jose, CA Aug 2023 - May 2025
Master of Science, Electrical Engineering GPA: 3.68/4.0
Coursework: Digital System Design and Synthesis, Logic Verification with UVM, SoC Design & Verification with System Verilog, ASIC CMOS Design, Advanced Computer Architectures

Gujarat Technological University, Ahmedabad, India July 2017 - May 2021
Bachelor in Engineering, Electronics and Communication GPA: 3.78/4.0
Coursework: VLSI Technology and Design, Microprocessor and Interfacing, Digital Electronics, Digital and Analog Circuit Design

SKILLS

Technical Skills: Functional Verification, System Verilog Assertion (SVA), Cover Groups, ASIC Design, Debugging

Languages and Methodologies: Verilog, System Verilog, UVM, Python, C

Tools: Synopsys (VCS, Design Compiler, Verdi, VC Formal, Prime Time), Xilinx Vivado, ModelSim, MATLAB, GVim

Concepts: Constraint Random Methodology, Formal Verification, Object Oriented Programing (OOPs), Computer Architecture

Bus Protocol: PCIe, AHB, APB, AXI

Soft Skills: Strong Communication and Writing Skills, Ability to work Independently and in Team, Goal Oriented, Problem Solving

PROJECTS

OFDM Communication System Verification (System Verilog, UVM, VCS) Mar 2024 - May 2024

- Developed UVM-based predictive testbench with constrained randomization to verify 128-point FFT/IFFT block for fixed-point OFDM processing and identifying scaling mismatches
- Created reusable verification environment with agents, drivers, sequencers, monitors, scoreboards, achieving 97% functional coverage
- Automated regression testing and debug workflows, reducing verification turnaround time by 25%

Vending Machine Functionality Verification (SystemVerilog, UVM, VCS) Feb 2024 - Feb 2024

- Developed UVM-based testbench and test plan to validate vending machine functionality, ensuring 100% correctness for exact, excess, and insufficient payment scenarios
- Created reusable UVM components like scoreboards, drivers, agents, monitors, virtual interfaces, enabling modular testbench design and faster debugging
- Implemented functional and code coverage models, achieving >95% coverage to drive verification completion

FPGA-Based Image Processing System Function Implementation (Verilog, Vivado) Oct 2023 - Dec 2023

- Implemented convolution-based image processing filters like edge detection, smoothing, Gaussian blur, box blur on PYNQ FPGA, achieving real-time 30 fps performance on 512×512 images
- Designed pipelined datapath components like MAC units, line buffers, image controllers to utilize parallelism, improving throughput by 40% compared to sequential design
- Validated system using Vivado simulations and FPGA testing, ensuring functional correctness across multiple image datasets

High-Performance AI Calculation Engine Design on SoC (SystemVerilog) Sept 2024 - Dec 2024

- Integrated floating-point multiplier and adder blocks with APB register interface and high-speed bus, improving memory bandwidth efficiency by 20% and enabling smooth SoC communication
- Developed synthesizable SystemVerilog RTL and met timing closure at 500 MHz, ensuring the design was fully compatible with performance targets
- Ran post-synthesis simulations and resolved timing issues, reducing verification turnaround, ensuring reliable functionality

WORK EXPERIENCE

Teaching Associate, SJSU Aug 2024 - May 2025

- Taught and mentored 50+ students in Continuous and Discrete-Time Systems lab, strengthening their understanding of DSP concepts
- Guided students in using LTspice and Python to model and analyze signals, circuits, and systems

Technical Student Assistant, SJSU June 2024 - May 2025

- Assisted with FPGA lab setups and hardware debugging using oscilloscopes, logic analyzers, spectrum analyzers, and SMD soldering to support digital design experiments